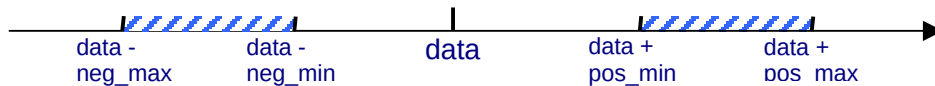


GEN_DELTA

Overview

gen_delta allows users to change the value of a multibit signal within a pre-defined values range. The delta, which defines the change of the input value, is calculated every time an input data signal changes its value. Four parameters (neg_min, neg_max, pos_min, pos_max) constrain the randomization of delta. The following figure illustrates the randomization constraints:



As it is illustrated in the figure above, the changed value for the data is randomly chosen within specified ranges. The randomization constraint enables users to change the value of an input data signal within a specified range of values.

Syntax

```
gen_delay #(min_clocks, max_clocks, bit_width) DelayInst (clk, rst_n,  
data_in, data_out);
```

Parameters	
Neg_Min_Delta	Minimum limit of negative value change
Neg_Max_Delta	Maximum limit of negative value change
Pos_Min_Delta	Minimum limit of positive value change
Pos_Max_Delta	Maximum limit of positive value change
bit_width	Data bit width. Default: 1
Interface	
Clk	Sampling clock for data
rst_n	Active-low reset
data_in	Input data signal. Bit width of this signal is defined with a "bit_width" parameter.
data_out	Output data signal. Bit width of this signal is defined with a "bit_width" parameter.

Verilog example

The current example shows how an 8-bit data_in signal is delayed by the random number of clock cycles [from 3 up to 6]

```
module test;
reg clk;

reg [7:0] data_in;
wire [7:0] data_out;
integer i;

always #15 clk <= ~clk;
gen_delay #(3,6,8) delay (clk, 1'b1, data_in, data_out);

initial begin
    $dumpvars;
    clk = 1;
    data_in = 0;

    for (i=0; i<200; i=i+1) begin
        @(posedge clk);
        if (i%10 == 0) data_in = i;
        else data_in = 0;
    end
    #100 $finish;
end
endmodule
```